TAHSIN AHMED FAHIM

Computer Science & Engineering Undergraduate | Full-Stack Developer | AI/ML Researcher

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PROFILE SUMMARY

Computer Science and Engineering undergraduate at AIUB (CGPA 3.90/4.00) focused on backend and full-stack development: REST APIs and web applications with NestJS, Next.js, TypeScript and PostgreSQL, and PHP/MySQL web applications, with additional projects in C#, Java and C++. Author of 3 IEEE conference papers on explainable machine learning.

EDUCATION

American International University-Bangladesh (AIUB) Dhaka, Bangladesh
B.Sc. in Computer Science and Engineering | Major: Software Engineering **CGPA: 3.90 / 4.00**
Expected Graduation: MAY, 2027
Relevant Coursework: Advanced Web Technology, OOP, Data Structures & Algorithms, Database Systems, Software Requirements Engineering, Software Quality & Testing, Machine Learning

BPATC School & College | Higher Secondary Certificate (HSC), Science **GPA: 5.00 / 5.00**

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, PHP, C++, SQL | Familiar: C#, Java, Python
Backend: NestJS, Node.js, REST APIs, JWT Authentication, PHP (MVC)
Frontend: Next.js, React, Tailwind CSS, HTML, CSS, AJAX
Databases: PostgreSQL, MySQL, Microsoft SQL Server, TypeORM
Core Concepts: OOP, Data Structures & Algorithms, MVC, Role-Based Access Control, Relational Database Design
ML & Research: Random Forest, XGBoost, SVM, SMOTE, SHAP, Probability Calibration
Tools & Desktop: Git, GitHub, VS Code, Visual Studio, LaTeX, .NET (Windows Forms), OpenGL / GLUT

FEATURED PROJECTS

PawsHome BD | Multi-Role Pet Adoption Platform ***NestJS, Next.js, TypeScript, TypeORM, PostgreSQL*** GitHub
- Built a platform where listers post pets, admins verify and approve them, adopters adopt and vets provide care.
- Built a NestJS REST API with 74 endpoints across 4 roles: JWT auth, bcrypt, DTO validation, mailer and Multer uploads.
- Designed the PostgreSQL schema with TypeORM relations; built the 41-page Next.js frontend with a dashboard per role.

Travel Management System ***PHP (MVC), MySQL, JavaScript (AJAX)*** GitHub
- Built a travel platform: scouts submit destinations, admins approve, users browse with AJAX search, filters and wishlists.
- 6-table MySQL schema; secured with CSRF tokens, prepared statements, hashed passwords and Remember Me login.

Cafe La Vien | Cafe Management System ***C#, .NET Windows Forms, SQL Server*** GitHub
- Built a desktop app with Admin (users, menu, sales) and Cashier (orders, cart, invoices) workflows, stock checks and VAT billing.

Bank Management System ***Java, OOP***
- Built deposits, withdrawals and inter-account transfers with a transaction history of IDs, dates, amounts, senders and receivers.

A Day in Winter: Rural Village Life ***C++, OpenGL, GLUT*** GitHub
- Built four animated OpenGL scenes (~2,000 lines) with modular drawing functions, timer animation and keyboard/menu control.

RESEARCH & PUBLICATIONS

Machine Learning Research | *Undergraduate Researcher, Department of Computer Science, AIUB*
- **Phishing detection:** 7 classifiers on 1,082,664 URLs from 8 sources; Random Forest best, SHAP explanations, McNemar's test.
- **Stroke risk:** LR, RF, XGBoost and SVM on imbalanced clinical data (SMOTE), with SHAP and Platt / isotonic calibration.
- **Agentic AI:** built a predictive framework for balancing autonomy and human oversight in agentic AI adoption.

Publications

1. **T. A. Fahim**, F. Mahmud, S. J. Snegdha, R. A. M. Rudro, T. Sayeed, and K. Nur, "Phishing URL Detection Using Machine Learning: A Comparative Study with SHAP Based Explainability on a Large-Scale Multi-Source Dataset," *IEEE ICACT 2026*, Kathmandu, Nepal.
2. S. J. Snegdha, **T. A. Fahim**, R. A. M. Rudro, T. Sayeed, and K. Nur, "Balancing Autonomy and Oversight: A Predictive Framework for Agentic AI Adoption," *IEEE ICACT 2026*, Kathmandu, Nepal.
3. **T. A. Fahim**, S. J. Snegdha, T. Rahannum, R. A. M. Rudro, and K. Nur, "Explainability and Calibration of Stroke Risk Prediction Models Based On AI," *IEEE COMPAS 2026*, University of Dhaka, Bangladesh (accepted).

ACHIEVEMENTS

- **Finalist,** Three Minute Thesis (3MT) Competition and Sciblitz 2.0 Science & Tech Contest, Chittagong University of Engineering & Technology (CUET)